A rank-one counterexample to the graphon Perron converse

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Lane 10 — candidate result for expert review

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Status. Candidate counterexample to the literal open problem in Bonnet–Pouradier Duteil–Sigalotti, together with an exact characterization inside the full-support rank-one class. The proof is elementary and appears complete, but the interpretation caveat in Section 5 is important.

1 Source question

Bonnet, Pouradier Duteil, and Sigalotti ask on printed page 25 of arXiv:2111.03900, after Theorem 4.7, whether the finite-dimensional Perron converse persists for graphon Laplacians [1]. Their question is:

> Is it true that an interaction topology is a countable disjoint union of strongly connected components whenever there exists an element $v \in L^\infty(I, \mathbb{R}_+^*)$ such that $\mathbb{L}^* v = 0$?

and we will see in Section 4.2 that this quantity does allow to quantify consensus formation for strongly connected topologies. Therein, we will also prove that the rescaled Laplacian operator is indeed a positive semi-definite operator over $L^2(I, \mathbb{R}^d)$.

Open problem. *Is the converse implication of Theorem 4.1 still valid in the context of graphon models described by Theorem 4.7? Namely, is it true that an interaction topology is a countable disjoint union of strongly connected components whenever there exists an element $v \in L^\infty(I, \mathbb{R}_+^*)$ such that $\mathbb{L}^* v = 0$?*

4.2 Exponential consensus formation for algebraically persistent topologies

In this section, we investigate the asymptotic formation of L^2-consensus for graphon models whose

Figure 1: The open problem in the source paper, printed page 25. The crop is rendered from the local copy `source_paper.pdf`.

Here $I = [0, 1]$, an adjacency kernel $a \in L^\infty(I \times I, [0, 1])$ defines

$$(\mathcal{A}x)(i) = \int_I a(i,j)x(j)\,dj, \qquad d(i) = \int_I a(i,j)\,dj, \qquad \mathbb{L} = M_d - \mathcal{A}.$$

The source's Definition 4.5 requires a strongly connected topology to satisfy both qualitative reachability through row supports and the quantitative bound $\inf_{i \in I} d(i) > 0$. A disjoint union must consist of at most countably many closed positive-measure components and satisfy the corresponding normalized degree bound with one uniform positive constant.

2 Proof intuition

The adjoint-null equation is a balance law, whereas Definition 4.5 also asks for a uniform interaction rate. Rank-one kernels separate these two demands completely. If $a(i,j) = p(i)q(j)$, the adjoint equation can be solved explicitly: the stationary weight is proportional to q/p. In contrast, the degree is proportional to p. Taking $p = q$ makes the stationary weight constant no matter how close p comes to zero. Full support prevents a partition from hiding the low-degree region inside different components.

3 Rank-one characterization

All assertions below are understood up to null sets, as is standard for L^∞ kernels. For the final continuous counterexample, ordinary and essential supports and infima give the same conclusion.

Theorem 1 (Full-support rank-one class)**.** *Let $p, q \in L^\infty(I)$ satisfy $0 < p, q \le 1$ almost everywhere, and put*

$$a(i,j) = p(i)q(j), \qquad Q = \int_I q(j)\,dj.$$

Then:

1. *A function $v \in L^\infty(I)$ with $v > 0$ almost everywhere satisfies $\mathbb{L}^*v = 0$ if and only if*

$$v = c\,\frac{q}{p}$$

for some $c > 0$. Consequently such a bounded strictly positive weight exists exactly when $q/p \in L^\infty(I)$; with $\int_I v = 1$ it is unique.

2. *If $q > 0$ almost everywhere, every nonzero row has support I. Hence a Definition 4.5 disjoint-union decomposition exists if and only if the whole space is its single component, which occurs if and only if*

$$\operatorname*{ess\,inf}_{i\in I} p(i) > 0.$$

In particular, bounded positive adjoint-null weights do not imply the quantitative strong-connectivity conclusion.

Proof. For $x \in L^2(I)$,

$$(\mathcal{A}x)(i) = p(i)\int_I q(j)x(j)\,dj, \qquad d(i) = Qp(i).$$

The adjoint of the rank-one operator is

$$(\mathcal{A}^*y)(i) = q(i)\int_I p(j)y(j)\,dj.$$

Writing $C = \int_I p(j)v(j)\,dj$, the equation $\mathbb{L}^*v = 0$ is therefore

$$Qp(i)v(i) = q(i)C \quad \text{for almost every } i.$$

Since $p, v > 0$ almost everywhere, $C > 0$, and this identity is equivalent to $v = (C/Q)(q/p)$. Conversely, if $v = cq/p$, then $C = c\int_I q = cQ$, so the displayed equation holds. Normalizing the integral fixes $c = (\int_I q/p)^{-1}$. This proves the first assertion.

For almost every i, $p(i) > 0$, so the row $a(i, \cdot)$ is a positive scalar multiple of q. Since $q > 0$ almost everywhere, its support is all of I. The qualitative path condition in Definition 4.5(a) is automatic: any two eligible points are joined in one step.

Suppose $\{I_n\}$ were a disjoint-union decomposition in the sense of that definition. Each I_n has positive measure. Choosing an eligible $i \in I_n$ gives

$$I = \operatorname{supp} a(i, \cdot) \subset I_n$$

up to null sets. Thus there can only be one component, namely I itself. On this component Definition 4.5(b) is precisely

$$0 < \inf_{i \in I} d(i) = Q \inf_{i \in I} p(i),$$

or, invariantly, $\operatorname{ess\,inf} p > 0$. The converse is immediate: when this bound holds, I itself satisfies both conditions of Definition 4.5. $\square$

4 The minimal counterexample

Corollary 2. *The open problem has a negative answer as literally stated. Define*

$$a(i, j) = ij \qquad (i, j \in [0, 1]).$$

Then $v \equiv 1$ belongs to $L^\infty(I, \mathbb{R}_+^*)$ and satisfies $\mathbb{L}^* v = 0$, but the topology is not a disjoint union of strongly connected components in the sense of Definition 4.5.

Proof. This is Theorem 1 with $p(i) = q(i) = i$ (the value at the null point $i = 0$ is immaterial). Directly,

$$d(i) = \int_0^1 ij \, dj = \frac{i}{2}, \qquad (\mathcal{A}1)(i) = \frac{i}{2},$$

and $\mathcal{A}$ is self-adjoint because a is symmetric. Hence $\mathbb{L}^* 1 = \mathbb{L} 1 = d - \mathcal{A} 1 = 0$. For each $i > 0$, the continuous function $j \mapsto ij$ has support I, so an admissible closed component containing i must be all of I. But

$$\inf_{i \in I} d(i) = \inf_{i \in I} \frac{i}{2} = 0,$$

contradicting Definition 4.5(b). $\square$

5 Interpretation caveat and corrected rank-one statement

The example is qualitatively strongly connected: for $i, j > 0$, the edge weight $a(i, j)$ is positive, and the exceptional point $i = 0$ is negligible. It fails only the uniform degree condition built into the source's terminology. Therefore:

- Under Definition 4.5 exactly as printed, Corollary 2 is a full counterexample.
- If "strongly connected components" in the open problem was intended to mean only qualitative mutual reachability, this example does not refute that weaker intended statement. It instead identifies the missing quantitative hypothesis.
- In the full-support rank-one class, the exact repair is $\operatorname{ess\,inf} p > 0$. The stationary equation alone controls q/p and cannot supply this bound.

There is also a same-source warning about novelty. In Section 5.4 the authors give a different symmetric, qualitatively connected kernel whose degree infimum is zero [1]. Symmetry automatically yields $\mathbb{L}^*1 = 0$. Thus the obstruction to the literal open problem is already latent in the source paper, although that section does not state that it answers the open problem. The rank-one calculation above isolates the mechanism and gives a complete elementary family, but priority for the bare negative answer should be treated as low.

6 Verification and novelty bounds

The proof is symbolic and uses no numerical computation. The following checks were made explicitly:

1. the adjoint of the rank-one operator was computed from its defining L^2 pairing;
2. substituting $v = cq/p$ reproduces the same scalar $C = cQ$;
3. full row support rules out every multi-component partition allowed by Definition 4.5;
4. for $a(i, j) = ij$, both the literal and essential degree infima vanish.

A bounded search on 9 August 2026 covered the run's registry, solution, attempt, and proof-gap indexes; arXiv id 2111.03900; the exact open-problem phrase; the paper title combined with "Perron"; and web searches for positive stationary graphon weights and strong components. It found the source and later citing papers, but no separate paper explicitly resolving this converse. This is not an exhaustive MathSciNet or zbMATH priority search. The latent Section 5.4 example is the principal novelty limitation.

Human-review recommendation. Check first whether the authors intended Definition 4.5 in full in the open problem. If so, the counterexample is immediate and the rank-one theorem is a sharp diagnosis. If only qualitative reachability was intended, retain the rank-one theorem as a statement correction and do not advertise the weaker problem as solved.

References

[1] B. Bonnet, N. Pouradier Duteil, and M. Sigalotti, *Consensus Formation in First-Order Graphon Models with Time-Varying Topologies*, Math. Models Methods Appl. Sci. 32 (2022), no. 11, 2121–2188; arXiv:2111.03900; DOI: 10.1142/S0218202522500518.